

Technology Terms Glossary

Simple definitions of common technology, AI, networking and cloud terms, written so a beginner can understand them. Each entry is self-contained.

Artificial Intelligence and Machine Learning Terms

Artificial Intelligence (AI)

Artificial Intelligence, or AI, means computer systems that do tasks which normally need human thinking, such as understanding language, recognising images or making decisions.

Machine Learning (ML)

Machine Learning is a part of AI where a computer learns patterns from examples instead of being given fixed rules. If you show a machine learning model thousands of cat photos, it learns to recognise cats.

Deep Learning

Deep Learning is machine learning that uses neural networks with many layers. Deep learning powers image recognition, speech recognition and modern language models.

Neural Network

A Neural Network is a model made of connected units called neurons, loosely inspired by the brain. Each connection has a weight, and training adjusts these weights so the network gives better answers.

Model

A Model in AI is the trained file that has learned patterns from data. You give the model an input and it produces an output such as a prediction or a sentence.

Training

Training is the process of showing a model many examples so it adjusts its internal numbers and gets better at a task. Training is usually slow and expensive.

Inference

Inference is using an already trained model to get an answer. When you ask a chatbot a question, that is inference. Inference is fast compared to training.

Dataset

A Dataset is the collection of data used to train or test a model. A dataset is usually split into a training set, a validation set and a test set.

Overfitting

Overfitting happens when a model memorises the training data instead of learning general patterns. An overfitted model scores well on training data but poorly on new data.

Underfitting

Underfitting happens when a model is too simple to capture the patterns in the data, so it performs badly on both training and new data.

Parameter

A Parameter is a number inside a model that is learned during training. When people say a model has 7 billion parameters, they mean it contains 7 billion of these learned numbers.

Hyperparameter

A Hyperparameter is a setting chosen by the engineer before training, such as learning rate or number of layers. Hyperparameters are not learned by the model itself.

Supervised Learning

Supervised Learning is training a model on data that already has correct answers, called labels. Spam detection trained on emails marked spam or not spam is supervised learning.

Unsupervised Learning

Unsupervised Learning is training a model on data without labels, so the model finds structure on its own. Grouping customers into segments is unsupervised learning.

Reinforcement Learning

Reinforcement Learning is training where an agent learns by trial and error, getting rewards for good actions and penalties for bad ones. Game-playing AI often uses reinforcement learning.

Natural Language Processing (NLP)

Natural Language Processing, or NLP, is the field of AI that deals with human language: translation, summarising, sentiment analysis and chatbots.

Computer Vision

Computer Vision is the field of AI that lets computers understand images and video, such as face recognition or detecting defects on a factory line.

Large Language Model (LLM)

A Large Language Model, or LLM, is an AI model trained on huge amounts of text that can read and write human language. Examples include GPT, Llama, Mistral and Claude.

Transformer

A Transformer is the neural network architecture behind almost all modern language models. Its key idea is attention, which lets the model weigh which words matter most.

Attention

Attention is a mechanism that lets a model focus on the most relevant parts of the input when producing each part of the output.

Token

A Token is a small piece of text, roughly a word or part of a word. Language models read and write in tokens, and pricing and context limits are measured in tokens.

Context Window

The Context Window is how much text a language model can consider at once, measured in tokens. If your input is longer than the context window, older text is cut off.

Prompt

A Prompt is the instruction or question you give to a language model. Better prompts usually produce better answers.

Prompt Engineering

Prompt Engineering is the practice of writing and refining prompts so a model gives accurate and useful answers, using techniques like giving examples and asking for step-by-step reasoning.

Hallucination

Hallucination is when a language model produces information that sounds confident but is wrong or invented. Retrieval Augmented Generation is one common way to reduce hallucination.

Fine-tuning

Fine-tuning is taking an already trained model and training it a bit more on your own specific data so it fits your task or style.

RAG (Retrieval Augmented Generation)

RAG, short for Retrieval Augmented Generation, is a technique where you first search your own documents for relevant text and then give that text to a language model along with the question. RAG lets a model answer from private or recent information it was never trained on, and it reduces hallucination.

Embedding

An Embedding is a list of numbers that represents the meaning of a piece of text. Texts with similar meaning have embeddings that are close together, which is how semantic search works.

Vector Database

A Vector Database stores embeddings and finds the closest matches quickly. FAISS, Chroma, Qdrant, Milvus and Pinecone are common vector databases used in RAG systems.

Chunking

Chunking is splitting long documents into smaller pieces before storing them in a vector database, so that search returns a focused passage instead of a whole book.

Semantic Search

Semantic Search finds results by meaning rather than exact keywords. Searching for 'how to fix slow internet' can return a document about bandwidth troubleshooting even without matching words.

Ollama

Ollama is a tool that lets you download and run open language models such as Llama, Mistral, Phi and Gemma on your own computer, with a simple command line and a local API. Ollama is popular for building private chatbots because the data never leaves your machine.

LangChain

LangChain is a framework that helps developers connect language models to data sources, tools and memory, and is often used to build RAG applications and agents.

Hugging Face

Hugging Face is a platform that hosts open machine learning models, datasets and libraries such as Transformers. It is a common starting point for AI engineers.

Agent

An AI Agent is a system where a language model can decide to use tools, take multiple steps, and act towards a goal rather than just answering once.

MLOps

MLOps is the practice of running machine learning systems in production reliably, covering deployment, monitoring, versioning and automatic retraining.

GPU

A GPU, or Graphics Processing Unit, is a chip that can do many calculations in parallel. GPUs make training and running AI models far faster than ordinary CPUs.

API

An API, or Application Programming Interface, is a defined way for one program to ask another program for something. AI models are usually made available to applications through an API.

Networking Terms

Network

A Network is a group of connected computers and devices that can share data with each other.

LAN

A LAN, or Local Area Network, connects devices in one small area such as a home, office or building.

WAN

A WAN, or Wide Area Network, connects networks across large distances, such as linking branch offices in different cities. The internet is the largest WAN.

IP Address

An IP Address is the unique number that identifies a device on a network, for example 192.168.1.10. It works like a postal address for data.

IPv4 and IPv6

IPv4 is the older addressing system using numbers like 192.168.1.1 and it is running out of addresses. IPv6 is the newer system with far more addresses, written in longer hexadecimal form.

Subnet and Subnetting

A Subnet is a smaller network carved out of a larger one. Subnetting is the skill of dividing IP address ranges into these smaller networks, and it is a core interview topic for a Network Engineer.

Router

A Router is a device that sends data between different networks and chooses the best path for it. Your home router connects your home network to the internet.

Switch

A Switch is a device that connects devices inside the same network and forwards data only to the correct port, unlike an older hub which sent it everywhere.

Firewall

A Firewall is a security device or software that allows or blocks network traffic based on rules, protecting a network from unwanted access.

DNS

DNS, the Domain Name System, translates human names like google.com into IP addresses that computers use. DNS is often called the phonebook of the internet.

DHCP

DHCP, the Dynamic Host Configuration Protocol, automatically gives devices an IP address when they join a network, so nobody has to set it by hand.

NAT

NAT, or Network Address Translation, lets many devices on a private network share one public IP address when reaching the internet.

VPN

A VPN, or Virtual Private Network, creates an encrypted tunnel over the internet so remote users can safely reach a private network.

VLAN

A VLAN, or Virtual LAN, splits one physical switch into separate logical networks, so departments can be isolated without extra hardware.

OSI Model

The OSI Model describes networking in seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application. Network Engineers use it to describe where a problem sits.

TCP/IP

TCP/IP is the set of protocols that runs the internet. TCP makes sure data arrives complete and in order, while IP handles addressing and routing.

TCP vs UDP

TCP is reliable and checks that every packet arrives, which suits web pages and file transfer. UDP is faster but does not guarantee delivery, which suits video calls and online games.

Port

A Port is a number that identifies a specific service on a device. Port 80 is HTTP, port 443 is HTTPS, port 22 is SSH.

Packet

A Packet is a small block of data sent across a network. Large files are broken into packets and reassembled at the destination.

Bandwidth

Bandwidth is how much data a network connection can carry per second, usually measured in Mbps or Gbps.

Latency

Latency is the delay before data starts arriving, usually measured in milliseconds. Low latency matters for gaming and video calls.

Routing Protocol

A Routing Protocol is how routers share information about paths. OSPF is common inside an organisation and BGP is the protocol that connects the internet.

Load Balancer

A Load Balancer spreads incoming traffic across several servers so no single server is overloaded and the service stays available.

Proxy Server

A Proxy Server sits between a user and the internet, forwarding requests. Proxies are used for filtering, caching and hiding internal addresses.

Cloud Computing

Cloud Computing means renting computing power, storage and services over the internet instead of owning physical servers. AWS, Microsoft Azure and Google Cloud are the main providers.

Cloud, DevOps and General Technology Terms

IaaS, PaaS and SaaS

IaaS gives you raw servers and storage. PaaS gives you a platform to run your code without managing servers. SaaS gives you ready-made software such as Gmail. The more you move from IaaS to SaaS, the less you manage yourself.

Virtual Machine

A Virtual Machine is a software computer running inside a physical computer, with its own operating system, letting one server run many isolated systems.

Container

A Container packages an application together with everything it needs to run, so it behaves the same on any machine. Containers are lighter and faster to start than virtual machines.

Docker

Docker is the most widely used tool for building and running containers.

Kubernetes

Kubernetes is a system that runs and manages many containers across many machines, handling scaling, restarts and networking automatically.

CI/CD

CI/CD stands for Continuous Integration and Continuous Deployment. It means code is automatically tested and released whenever a developer pushes changes, which reduces manual mistakes.

Git

Git is a version control system that tracks changes to code and lets many developers work on the same project without overwriting each other. GitHub and GitLab are websites that host Git repositories.

Infrastructure as Code (IaC)

Infrastructure as Code means describing servers and networks in text files, for example with Terraform, so the setup is repeatable and reviewable instead of clicked by hand.

Microservices

Microservices is a design where an application is split into many small independent services that talk over APIs, instead of one big program called a monolith.

REST API

A REST API is a common style of web API that uses standard HTTP methods such as GET to read data and POST to send data, usually exchanging JSON.

JSON

JSON, or JavaScript Object Notation, is a simple text format for structured data that both humans and programs can read easily. Most APIs send and receive JSON.

SQL

SQL, or Structured Query Language, is the language used to store and retrieve data in relational databases such as MySQL and PostgreSQL. SQL is required for almost every data role.

NoSQL

NoSQL databases store data in flexible formats such as documents or key-value pairs instead of fixed tables. MongoDB and Redis are examples.

ETL

ETL stands for Extract, Transform, Load. It is the process of pulling data from sources, cleaning it, and loading it into a warehouse for analysis.

Data Warehouse

A Data Warehouse is a central database designed for analysis and reporting over large amounts of historical data. Snowflake, BigQuery and Redshift are examples.

Agile and Scrum

Agile is a way of working where software is built in short cycles with frequent feedback. Scrum is a popular Agile method using sprints, daily stand-ups and a product backlog.

Open Source

Open Source software has publicly available source code that anyone can read, use and improve. Linux, Python and many AI models are open source.

Linux

Linux is a free open-source operating system that runs most servers, cloud systems and network devices. Comfort with the Linux command line is expected in most infrastructure roles.

SLA

An SLA, or Service Level Agreement, is a promise about service quality, such as guaranteeing that a system will be available 99.9 percent of the time.

Encryption

Encryption scrambles data so that only someone with the correct key can read it. HTTPS uses encryption to protect data travelling between your browser and a website.

Phishing

Phishing is an attack where a fake email or message tricks a person into giving away passwords or clicking a harmful link. It is one of the most common causes of security breaches.

Zero Trust

Zero Trust is a security approach that assumes no user or device is automatically trusted, so every request must be verified even inside the company network.